Critical Thinking

The following activities give you critical thinking challenges. Challenge labs give you an opportunity to use the skills you have learned to perform a task without step-by-step instructions. Case projects offer a practical networking setup for which you supply a written solution.

Challenge Lab 4-1: Creating a 1000BaseT Crossover Cable

Time Required: 20 minutes

Objective: Create a 1000BaseT crossover cable.

Required Tools and Equipment: Wire cutter and cable stripper, RJ-45 crimping tool, 2 to 4 feet of Cat 5e or Cat 6 cable, two RJ-45 plugs, and two devices that support 1000BaseT (which can be two PCs with 1000BaseT NICs or two 1000BaseT switches)

Description: In this challenge lab, you create a crossover cable that supports 1000BaseT Ethernet and test it between two devices. Then you verify that the devices connect at 1000BaseT speed.

- How is a 1000BaseT crossover cable different from a 100BaseT crossover cable?

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- How did you verify that the devices connected at 1000BaseT?

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Challenge Lab 4-2: Creating a Rollover Cable

Time Required: 20 minutes

Objective: Create a rollover cable.

Required Tools and Equipment: Wire cutter and cable stripper, RJ-45 crimping tool, 2 to 4 feet of Cat 5e or Cat 6 cable, two RJ-45 plugs, a Cisco-managed device (such as a switch or a router with a console port), a PC with terminal emulation software installed (such as PuTTY), and a DB-9-to-RJ-45 adapter

Description: In this challenge lab, you create a rollover cable, using the DB-9-to-RJ-45 adapter to connect the PC's DB-9 serial port to one end of the rollover cable and the other end to the device's console port. If the PC doesn't have a DB-9 serial port, USB-to-RJ-45 serial port adapters are available. Run PuTTY to connect to the managed switch or router's console. You might need to research the PuTTY settings required to make the connection.

- Why might you need a rollover cable?

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- What's the pinout for a rollover cable?

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Case Project 4-1

During the design of most real-world networks, you'll discover that using more than one type of networking medium is common. The usual reasons for needing more than one type of medium include the following:

- Two or more areas must be interconnected, and the distance separating them is greater than the maximum segment length for the type of medium used in (or best suited for) each area.
- A connection must pass through a high-interference environment (across some large transformers, near heavy-duty electrical motors, and so on). Failure to use a different type of medium increases the risk of impeding data flow. This reason is especially common for choosing fiber-optic cable or wireless in many networks, particularly when connecting floors in an office building and the only available pathway is the elevator shaft.
- Certain parts of an internetwork might have to carry more traffic than other parts. Typically, the segment where traffic aggregates is the backbone, a common cable segment that interconnects subsidiary networks. (Think of a tree trunk as the backbone and its major branches as cable segments.) Often, a higher-capacity cable is used for a backbone (for example, fiber-optic cable or Cat 6 cable rated for Gigabit Ethernet), along with a higher-speed networking technology for attachments to the backbone. This arrangement means outlying segments might use conventional 10 or 100 Mbps Ethernet, and the backbone uses 1 Gbps or 10 Gbps Ethernet.

Using this information, suggest solutions that involve at least two types of networking media, if possible, to address the following problems:

- A—XYZ Corp. is planning a new network. Engineers in the design shop must have connections to accountants and salespeople in the front office, but all routes between the two areas must traverse the shop floor, where arc welders and metal-stamping equipment create potent amounts of EMI and RFI. Given that both the design shop and front office use 10BaseT (twisted-pair Ethernet), how might you interconnect these two areas? What medium guarantees immunity from interference?
- B—After the front-office network at XYZ Corp. is set up, an accountant realizes that if the loading dock connected to the network, dock workers could log incoming and outgoing shipments and keep the inventory more current. Even though the loading dock is nowhere near the shop floor, the dock is 1100 feet from the front office. What kinds of cable will work to make this connection? What kind would you choose and why?
- C—ABC Company occupies three floors in a 10-story building, where the elevator shaft provides the only path to all these floors. In addition, users on the 9th and 10th floors must access a collection of servers on the 8th floor. Explain what kind of connections would work in the elevator shaft. If more than one choice is possible, pick the best option and explain the reasons for your choice. Assuming that interfloor connections might someday need to run at much higher speeds, reevaluate your choice. What's the best type of medium for open-ended bandwidth needs? Explain your answer.

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Case Project 4-2

XYZ Corp.'s facilities in Nashua, New Hampshire, are two office buildings 400 feet apart, each with its own LAN. To connect the two networks, you plan to dig a trench and lay cable in conduit between the two buildings. You want to use fiber-optic cable, but your budget-conscious facilities manager wants to use 100 Mbps Ethernet over twisted-pair cable. Which of the following reasons can you use to justify fiber-optic cable in this case, and why?

- a: Twisted pair won't span a 400-foot distance.
- b: Fiber-optic cable is cheaper and easier to work with than twisted pair.
- c: Twisted pair is a conductive cable and can therefore carry current based on the difference in ground potential between the two buildings.
- d: Fiber-optic cable leaves more room for growth and future needs for increased bandwidth than twisted pair does.

Case Project 4-3

TVBCA has just occupied a historic building in downtown Pittsburgh where 15 employees will work. Because of codes for historic buildings, TVBCA isn't permitted to run cables inside walls or ceilings.

Required result: Employees must be able to share files and printers, as in a typical LAN environment, without using cables.

Optional desired results: Employees must be able to use their laptops or tablets and move freely throughout the office while maintaining a network connection. Because of the size of some computer-aided design (CAD) files that employees often use, data transfer speeds should be at least 100 Mbps and the connection should be secure.

Proposed solution: Install an 802.11ac wireless access point and configure each mobile device to connect to the AP with WPA2 encryption. Which of the following results does the proposed solution deliver? Explain your answer.

- a: The proposed solution delivers the required result and both optional desired results.
- b: The proposed solution delivers the required result and only one of the two optional desired results.
- c: The proposed solution delivers the required result but neither optional desired result.
- d: The proposed solution does not deliver the required result.

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